

COGNODB // STARTUP TEAM MATCHING GRAPH

Complete System Overview, Architecture, Cypher Graph Engine & Feature Guide

Graph DB: Neo4j / CognoDB Bolt Protocol | Format: Declarative Cypher | Version: 1.0.0

1. EXECUTIVE SUMMARY & PURPOSE

CORE OVERVIEW

The CognoDB Startup Team Matching Graph is a full-stack, graph-native intelligence platform engineered for tech ecosystems, venture studios, and talent networks. Traditional relational databases (SQL) struggle with recursive joins and multi-hop social connectivity, degrading exponentially as network depth increases. This platform leverages native property graph traversal (Neo4j / CognoDB) using Cypher queries to discover optimal team compositions, compute warm introduction paths, and model multi-dimensional relationships in sub-millisecond latencies.

2. GRAPH TOPOLOGY & DATA MODEL (PROPERTY GRAPH)

SCHEMA SPEC

The system models a 4-entity property graph with weighted and contextual relationships:

(:Person)	Candidates, founders, engineers, and designers with title, experience, location, and avatar.
(:Skill)	Technical & domain competencies (e.g. PyTorch, Distributed Systems, Rust, Product Design).
(:Project)	Previous ventures, open-source repos, and enterprise apps with completion status.
(:Company)	Current and past employers across AI, FinTech, Autonomous Tech, and Cloud.

[:KNOWS]	(Person -> Person)	Colleague, co-founder, or advisor connection with relationship context.
[:HAS_SKILL]	(Person -> Skill)	Skill proficiency (Expert, Advanced, Intermediate) with years of experience.
[:WORKED_ON]	(Person -> Project)	Project contribution with specific role (Lead Architect, Maintainer, etc.).
[:WORKS_AT]	(Person -> Company)	Employment history and past corporate tenures.

3. APPLICATION FEATURES & CAPABILITIES

FEATURE BREAKDOWN

A. Interactive Network Graph View & Topology Explorer

- * Dynamic 2D Canvas rendering engine with high-DPI retina display support.
- * Zero-idle CPU footprint: Synchronous relaxation layout calculations settle in <3ms, stopping battery drain.
- * Interactive node dragging, real-time panning, smooth wheel-zooming, and focal center re-anchoring.
- * Entity filter chips to dynamically toggle visibility of People, Skills, Projects, and Companies.
- * Inspector Drawer providing instant 1-hop attribute inspection and direct profile routing.

B. Shortest Path & Warm Introduction Traversal Engine

- * Executes Cypher `shortestPath((a:Person)-[:KNOWS*..6]-(b:Person))` across graph topologies.
- * Calculates exact degrees of separation (1st, 2nd, 3rd, 4th degree).
- * Extracts edge contexts along the chain (e.g. "Former co-founders at Voxel", "Stanford AI Lab").
- * Computes trust scores, common mutual colleagues, and structured referral outreach drafts.

C. Algorithmic Startup Team Matcher

- * Graph-pattern matching algorithm to assemble balanced multi-disciplinary teams.
- * Scans candidates with complementary skill matrices (Frontend, Backend, AI/ML, Design, Product).
- * Prioritizes historical working chemistry (candidates sharing past companies or projects).
- * Minimizes social distance: Assembles teams whose members are already 1 or 2 hops apart in the graph.

D. Talent Graph Directory & Multi-Dimensional Search

- * Instant multi-filter search across skills, roles, companies, locations, and seniority.
- * Candidate cards with direct profile expansion detailing all graph edges and historical ventures.
- * One-click graph re-centering to inspect any candidate's immediate neighborhood.

E. Cypher Query Laboratory & SQL Equivalence Benchmark

- * Interactive Cypher REPL executing live declarative graph queries directly on CognoDB.
- * Dynamic JSON query parameter editor for testing parameterized queries (\$personId, \$skillName, etc.).
- * Head-to-head complexity analysis: Compares O(d) pointer dereferencing vs O(N^d) SQL recursive CTE joins.
- * Real-time execution latency tracking in milliseconds and detailed tabular result inspection.

F. Database Management & Schema Inspector

- * Live connection status indicator (Neo4j / CognoDB Bolt driver).
- * Real-time node and relationship counter displaying total active entities in the database.
- * One-click seed generator to populate or reset realistic Silicon Valley / AI talent network graphs.

4. WHY GRAPH DATABASES (CYPHER VS SQL CTEs)

PERFORMANCE COMPARISON

In relational databases, querying relationships across 2+ hops requires multiple recursive self-joins or WITH RECURSIVE Common Table Expressions (CTEs), resulting in exponential join overhead. Graph databases use index-free adjacency where each node points directly to its physical neighbors in memory:

```
// CYPHER (Declarative Graph Pattern)

MATCH path = shortestPath(
  (a:Person {id: $pA})-
  [:KNOWS*..6]-(b:Person {id: $pB})
)
RETURN path, length(path);

Time: 1.2ms | Complexity: O(d)
```

```
-- SQL (Recursive CTE & Self-Joins)

WITH RECURSIVE traverse(p1, p2, depth) AS (
  SELECT src, dst, 1 FROM knows WHERE src=$pA
  UNION
  SELECT k.src, k.dst, t.depth+1 FROM knows k
  JOIN traverse t ON k.src = t.dst
  WHERE depth < 6
)
SELECT t1.src, t1.dst FROM traverse t1 WHERE t1.p2=$pB;
```

5. ARCHITECTURE & TECHNOLOGY STACK

SYSTEM ARCH

[Frontend SPA]	React 19, Tailwind CSS v4, Lucide Icons, Canvas 2D Retina Acceleration
[Backend Server]	Node.js, Express.js API, RESTful Subgraph & Traversal Endpoints
[Graph Engine]	Neo4j Driver / CognoDB Bolt Protocol (Native Cypher Pattern Engine)
[Algorithms]	Shortest Path (Dijkstra / BFS), Bounded Subgraph Relaxation, Jaccard Skill Clustering